Metric Geometry Princeton Quals Questions

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Contents

Question 0.0. (Naor-2023) *What can you say about Euclidean sections of ℓ_1^n?*

Question 0.1. (Naor-2023) *What is Kashin's Theorem?*

Question 0.2. (Naor-2023) *Define cotype.*

Question 0.3. (Naor-2023) *Prove the bound $k(X) \gtrsim n$ for cotype 2.*

Question 0.4. (Naor-2023) *State and prove Bourgain's embedding theorem.*

Question 0.5. (Guth-2010∗) *What's the systolic inequality?*

Question 0.6. (Gabai-2010∗) *What's aspherical? What if the manifold is unbounded?*

Question 0.7. (Guth-2010∗) *Sketch a proof of the systolic inequality for the n-torus.*

Question 0.8. (Guth-2010∗) *Give an isoperimetric inequality for filling loops in the 3-manifold $S^2 \times \mathbb{R}$ where S^2 has the round unit sphere metric.*

Question 0.9. (Guth-2010∗) *Given two loops of length L_1, L_2, the distance between the closest points on two loops is $g = 1$, what's the maximum linking number?*